GENERALIZATIONS LIFT ALONG FLAT MORPHISMS OVER A UR-FINITE POINT

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Editorial note. The parenthetical on when the UR-finiteness hypothesis holds (decent, quasi-separated or representable X) is context from the cited section, not an obligation.

Allowed standard prerequisites (editorial).

- Specialization lifting. lemma-specialization (carded L26708). Lifts x' -> x to u' -> u on the etale cover U -> X.
- Points over fibre products. Properties of Spaces, lemma points-cartesian (cross-chapter, named). Provides z in U x_X Y over (u, y).
- Scheme case. Morphisms, lemma generalizations-lift-flat (cross-chapter, named). Handles the flat morphism of schemes U x_X Y -> U.

Source reference. Original lemma 03JV: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 732–732. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03K2: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1544–1544. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03HV: focused-seeds/geometry/S9-decent-generalizations-lift-flat/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 03I7: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1232–1232. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-decent-generalizations-lift-flat/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a flat morphism of algebraic spaces over S. Let $x, x' \in |X|$ and assume $x' \rightsquigarrow x$, i.e., x is a specialization of x'. Assume the pair (X, x') satisfies the equivalent conditions of Lemma 03JV (for example if X is decent, X is quasi-separated, or X is representable). Then for every $y \in |Y|$ with $f(y) = x$, there exists a point $y' \in |Y|$, $y' \rightsquigarrow y$ with $f(y') = x'$.*

Proof. (The parenthetical statement holds by the definition of decent spaces and the implications between the different separation conditions mentioned in Section 03I7.) Choose a scheme V and a surjective étale morphism $V \to Y$. Choose $v \in V$ mapping to y. Then we see that it suffices to prove the lemma for $V \to X$. Thus we may assume Y is a scheme. Choose a scheme U and a surjective étale morphism $U \to X$. Choose $u \in U$ mapping to x. By Lemma 03K2 we may choose $u' \rightsquigarrow u$ mapping to x'. By Properties of Spaces, Lemma 03H4 we may choose $z \in U \times_X Y$ mapping to y and u. Thus we reduce to the case of the flat morphism of schemes $U \times_X Y \to U$ which is Morphisms, Lemma 03HV. □